Docker Generation & Testing Manual

Container Architecture, Health Monitoring & Operational Verification

Framework: MASTER_SDLC_AGENTS | DevOps Manual | Date: 2026-10-23 | Status: Production-Ready

1. Container Architecture & Security Specs

• Base Image: python:3.10-slim (Minimal Debian footprint)
• System Packages: libgl1, libglib2.0-0, curl
• Security User: Non-privileged appuser (UID 1000) for unprivileged execution
• Service Port: 8501 (Mapped 1:1 on host)
• Health Check: curl -f http://localhost:8501/_stcore/health

2. Step-by-Step Container Deployment

1. Build and launch container in detached mode:
   docker compose up --build -d

2. Verify container running state:
   docker ps

3. Inspect live container logs:
   docker logs -f lane_detection_dashboard

4. Access web dashboard in browser: http://localhost:8501

3. Health Monitoring & Maintenance

Healthcheck Verification:
Docker executes automated probes every 30s. The status in 'docker ps' displays '(healthy)' after 20s initial boot.

Teardown:
docker compose down